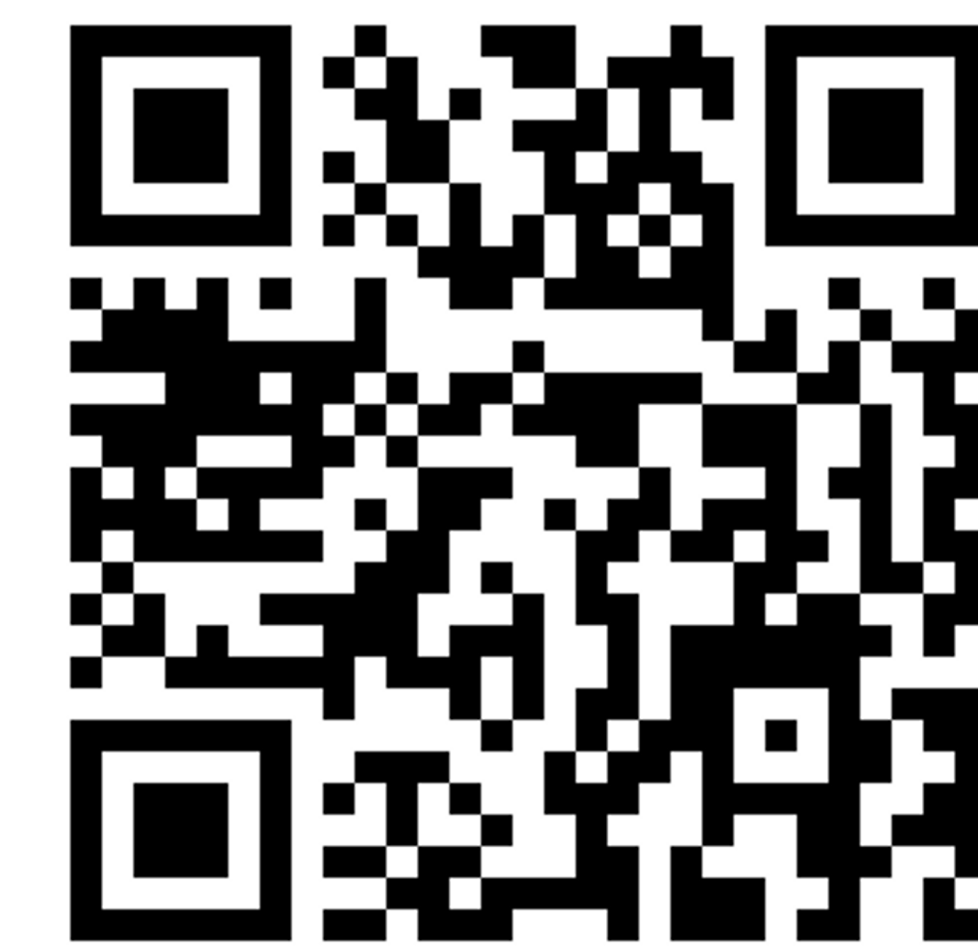


# PoSh: Using Scene Graphs to Guide LLMs-as-a-Judge for Detailed Image Descriptions

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## Motivation

- detailed image description is a useful, important task
  - measures *comprehensive* image understanding
  - better proxy for diverse, real-world user requests
  - accessibility applications (e.g. alt-text generation)
- but... evaluation is **hard**
  - human judgment is costly (~20 min per generation!)
  - most metrics were either designed for short texts or rely on calls to foundation models that are not replicable
  - moreover, long text requires error localization to disambiguate generations with similar overall scores & inform prompt / model improvements

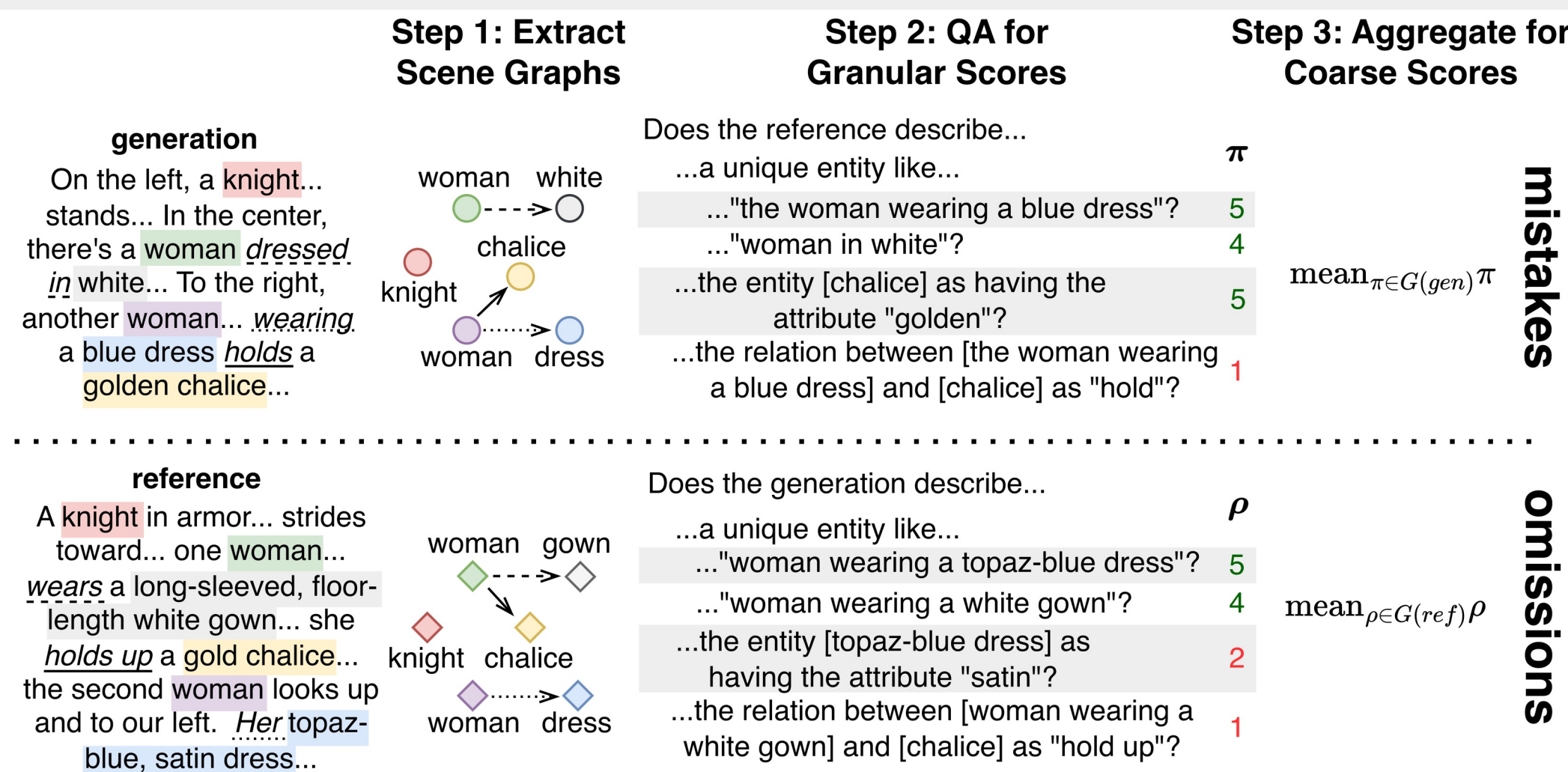


The Baptism of Christ (1535), Paris Bordone

## PoSh 🎩, a replicable, interpretable metric...



The Red Cross Knight (1793), John Singleton Copley



it outperforms existing metrics & GPT4o, it's robust to image type...

|              |    |                                                                                     |                                                                                     | DOCENT   |        |        |           |        |        |                 |        |        | CapArena |       |       |
|--------------|----|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------|--------|--------|-----------|--------|--------|-----------------|--------|--------|----------|-------|-------|
|              | Θ  |  |  | Mistakes |        |        | Omissions |        |        | Overall Quality |        |        | Desc     | Model |       |
|              |    |                                                                                     |                                                                                     | acc      | ρ      | τ      | acc       | ρ      | τ      | acc             | ρ      | τ      | acc      | ρ     | τ     |
| length       |    |                                                                                     | ✓                                                                                   | 30.5     | −0.270 | −0.206 | 37.8      | −0.002 | −0.001 | 38.0            | −0.160 | −0.121 | 58.7     | 0.710 | 0.582 |
| SPICE        | 32 | ✓                                                                                   | ✓                                                                                   | 41.3     | 0.308  | 0.234  | 55.0      | 0.464  | 0.360  | 58.5            | 0.458  | 0.349  | 41.7     | 0.275 | 0.231 |
| CAPTURE      |    | ✓                                                                                   | ✓                                                                                   | 43.3     | 0.259  | 0.194  | 53.8      | 0.447  | 0.340  | 56.0            | 0.453  | 0.347  | 52.5     | 0.613 | 0.538 |
| Qwen3        |    | ✓                                                                                   | ✓                                                                                   | 57.7     | 0.282  | 0.235  | 53.5      | 0.286  | 0.253  | 61.2            | 0.289  | 0.257  | 56.2     | 0.899 | 0.714 |
| LLaVa Critic |    | ✓                                                                                   | ✓                                                                                   | 62.8     | 0.412  | 0.351  | 57.0      | 0.509  | 0.430  | 66.8            | 0.546  | 0.461  | 64.0     | 0.987 | 0.934 |
| DCScore      |    | ✓                                                                                   | ✓                                                                                   | 62.8     | 0.541  | 0.422  | 54.0      | 0.395  | 0.298  | 62.8            | 0.471  | 0.362  | -        | -     | -     |
| GPT4o        |    | ✓                                                                                   | ✓                                                                                   | 58.5     | 0.484  | 0.396  | 56.0      | 0.380  | 0.303  | 67.3            | 0.510  | 0.402  | 55.4     | 0.890 | 0.802 |
| GPT5         |    | ✓                                                                                   |                                                                                     | 62.5     | 0.511  | 0.423  | 53.2      | 0.421  | 0.332  | 68.0            | 0.540  | 0.440  | 59.1     | 0.956 | 0.846 |
| PoSH         | 14 | ✓                                                                                   | ✓                                                                                   | 60.7     | 0.519  | 0.405  | 62.7      | 0.581  | 0.451  | 70.7            | 0.599  | 0.466  | 59.2     | 0.931 | 0.796 |

it's interpretable...

& it's a capable reward function!

|              | Random | 4GramEmbed | SGEmbed | PoSH         |
|--------------|--------|------------|---------|--------------|
| Mistakes F1  | 0.503  | 0.483      | 0.514   | <b>0.580</b> |
| Omissions F1 | 0.499  | 0.641      | 0.658   | <b>0.680</b> |

|                   | Mistakes | Omissions | Overall Quality |
|-------------------|----------|-----------|-----------------|
| PoSH-tuned vs SFT | -0.243   | 0.432     | 0.135           |

## Contributions

- PoSh, a new metric for evaluating detailed image descriptions
  - interpretable: coarse scores grounded in granular span localized scores (e.g. attribute mis-attachment)
  - replicable: entirely open-weight, inexpensive to use
  - state-of-the-art: outperforms existing metrics & GPT4o; robust to image type; a capable reward function
- DOCENT, a new detailed image description benchmark
  - visual art: complex scenes that stress generalization
  - 1,750 images with expert-written references from the US National Gallery of Art for detailed description evaluation
  - 900 granular & coarse judgments of generations from art history students for validating detailed description metrics

## DOCENT 🏛️, a new detailed image description benchmark

|  | DCI   | DOCCI | Compre | ImageIn | DeCap | Detail | Cap | DOCENT            |
|--|-------|-------|--------|---------|-------|--------|-----|-------------------|
|  | web   | web   | web    | web     | -     | web    | -   | art               |
|  | crowd | crowd | crowd  | crowd+  | -     | model  | -   | expert            |
|  | -     | -     | -      | -       | -     | coarse | -   | granular & coarse |



Osprey and Weakfish (1829), John James Audubon

| Reference                                                                                                                                                                                                                                                                         | Generation 1                                                                                                                                                                                                                                           | Generation 2                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A bird in flight with its hooked beak wide open, wings held high, holds a shimmering green and white fish in its talons in this vertical painting. The bird and fish fill the composition against a pale, hazy landscape... The bird surges forward, to our left, its head low... | The painting portrays a dynamic scene of a falcon in flight. The falcon, with its wings fully spread, is positioned in the upper left quadrant of the canvas...The falcon's head is turned slightly to the right, giving the impression of movement... | The painting depicts a majestic hawk in mid-flight, its wings fully extended against a light blue sky with wispy white clouds. The hawk's plumage is predominantly white, with dark brown wings and tail feathers featuring lighter brown stripes... |
| Coarse Judgments                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                      |
|                                                                                                                                                                                                                                                                                   | Mistakes: Generation 2 is much better                                                                                                                                                                                                                  | Missing Details: Generation 2 is slightly better                                                                                                                                                                                                     |
|                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                        | Overall: Generation 2 is much better                                                                                                                                                                                                                 |

Bright light from the upper left illuminates two muscular men and a winged angel against a dim landscape in this square painting. They all have pale skin, and the two men have dark brown hair and beards. One man, Jesus, stands ankle-deep in a rippling dark blue stream in the center of the composition facing us with downcast eyes. He is nude except for a white cloth that drapes over one shoulder and is tied around his hips. The other end of the cloth drops into the water. Jesus's hands are pressed together in prayer at chest-height closer to his right shoulder, to our left. He leans his torso in the other direction and looks down at the water. The second man stands on the riverbank to our right. His body faces our left but his left shoulder, closer to us, dips down so his broad back is angled toward us. His lower body is wrapped in a cobalt-blue garment bordered with what looks like sheepskin. His near hand swings back to clutch the trunk of a tree growing immediately behind him. In his other hand, he holds up a pewter-grey dish that dribbles water onto Jesus's head. On the left riverbank, the copper-haired angel looks on, facing our right in profile but the body faces us. The angel wears a sleeveless, pumpkin-orange robe with a shimmering salmon-pink mantle draped over the right shoulder, to our left. Petal-pink and dark grey wings extend from the shoulders, and the lower legs and feet are bare. The shadowy landscape has a dark green field stretching back to jagged, sapphire-blue mountains along the horizon, which comes four-fifths of the way up this composition. Pale blue clouds skim across the vivid blue sky, where three chubby, winged babies float. One holds a triangle and a rod with which to strike it, and another might hold a rectangular musical instrument. Blue clouds split nearby to show a sliver of golden yellow light, and possibly more angels.

entities, attributes and relations (and stopwords)